



Foundations of Finance



Cash Flows for Project Evaluation



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Problem Set 2: Key Results

- ▶ 1. Coupon bond \$1,017.47; $f_2 = 3.50\%$, $f_3 = 4.20\%$
- ▶ 2. Bank equity €32.05m $\rightarrow$ -€18.04m after +150 bp; 93 bp is enough to wipe it out
- ▶ 3. $V_0 = \$44.50 < \45.00 : overvalued ($7.78\% < 9\%$). Preferred: \$25.00; 6.82% at \$22
- ▶ 4. PVGO = 0 (ROE = k), +\$12.50 (ROE > k), -\$3.00 (ROE < k)
- ▶ 5. NPV A = 44.80, B = 71.03; IRR A = 14.14%, B = 13.33%. Take B. Project C: IRRs of 10% and 50%
- ▶ 6. $V_{15} = \$36.43$, $V_5 = \$23.21$, $V_0 = \$13.78$; 55% of the value is terminal value
- ▶ 7. The NPV profiles of A and B cross at 12.38%

Partial Review of the Last Class

- ▶ An assembly line earns \$5M a year after \$10M of depreciation, taxed at 20%. Why is its cash flow \$14M and not \$4M?
- ▶ Depreciation is not a cash flow. Why does it still change the cash flow?
- ▶ Why do we leave interest payments out of free cash flow?
- ▶ A firm's inventory rises by \$30M this year. What happens to its free cash flow?
- ▶ Apple's free cash flow is above its net income. Why?

Introduction

- ▶ Deciding whether to proceed with a project is different from valuing an entire company
 - ▶ Many parts of the company are unaffected by the project
 - ▶ Need to focus on the marginal change with vs without the project
 - ▶ Key concept: incremental cash flows
- ▶ Outline:
 - ▶ Using incremental cash flows for project decisions
 - ▶ Key issues in computing incremental cash flows
 - ▶ Mini Case: IM&C's Garden Fertilizer Project

Incremental Cash Flows

- ▶ A company should proceed with a project if

$$NPV(\text{Cash Flows With Project}) > NPV(\text{Cash Flows Without Project})$$

- ▶ We can rewrite this as

$$NPV(\text{Cash Flows With Project}) - NPV(\text{Cash Flows Without Project}) > 0$$

- ▶ Because NPV is additive, this is equivalent to

$$NPV(\text{Cash Flows With Project} - \text{Cash Flows Without Project}) > 0$$

- ▶ The net cash flows with vs without the project are called the incremental cash flows

Incremental Cash Flows

- ▶ The incremental cash flows method is based on two important ideas:
- ▶ Conceptual: change in cash flows matters, not the total
 - ▶ Companies are often still NPV-positive even if they add a negative NPV project
 - ▶ Important to compute the change in value from the project to make the correct decision
- ▶ Practical: incremental cash flows are much easier to forecast
 - ▶ In principle, could reach same decision by comparing NPV with and without project
 - ▶ But requires forecasting all cash-flow generating operations of the company
 - ▶ For large companies with multiple divisions, this is an enormous effort
 - ▶ With incremental cash flows, unaffected cash flow streams “cancel out,” usually leaving only a few key series to forecast

Incremental Cash Flows: Key Issues

- ▶ Treatment of sunk costs
- ▶ Allocated (overhead) costs
- ▶ Side benefits and costs
- ▶ Opportunity cost of unused resources
- ▶ Working capital requirements
- ▶ Effects of closure on cash flows

Sunk Cost + Allocated Overheads

- ▶ **Sunk costs:** any expenditure that has already been incurred or committed and cannot be reversed, even if the project is rejected
 - ▶ Sunk costs are not incremental $\Rightarrow$ should not be included
 - ▶ Market testing and R&D are often sunk by the time a go-ahead decision is made. How should these costs be considered in the capital allocation process within firms?
- ▶ **Allocated overhead costs:** expenses allocated to individual projects from a centralized pool (e.g., rent for corp. headquarters)
 - ▶ Should only be included if they would change because of the project
 - ▶ Note the difference between cost accounting and finance!

Side Benefits and Costs

- ▶ A project may provide benefits for other parts of the firm
 - ▶ Such benefits count as incremental cash flows for the project
- ▶ Ex: Starbucks' Pumpkin Spice Latte or “PSL”
 - ▶ *“The PSL isn’t just a drink—it’s a pop culture event. Starbucks’ clever marketing played a vital role, creating anticipation by teasing launch dates and harnessing the power of scarcity. Customers now count down the days to “PSL season.” Its limited-run nature drove urgency and repeated visits.... Not only does the PSL bring in direct sales, it also boosts traffic for baked goods, sandwiches, and other drinks—an effect the brand leverages across multiple menu categories each fall.”*
(Food and Beverage Magazine, 2025)
- ▶ Conversely, damaging or cannibalizing profits from another project should count against the incremental cash flows

Opportunity Costs

- ▶ An opportunity cost arises when a project uses a resource that is already owned by the firm, but could have instead been used in another project
- ▶ In this case, we should deduct the foregone benefit from the best alternative use of the resource
- ▶ Ex: A firm wants to use a warehouse it bought years ago for \$1 million for a new project. Similar warehouses currently rent for \$5,000 per month. What, if any, is the opportunity cost?
 - A. \$1 million
 - B. \$5,000 per month
 - C. \$0

Working Capital + Closure

- ▶ A project may require an increase in working capital to support its cash flow cycle
 - ▶ As discussed earlier, changes in NWC should be subtracted from free cash flows
 - ▶ Working capital is generally recovered at the end of the project
- ▶ Effects of closure on cash flows
 - ▶ Assets may be liquidated at the end of a project
 - ▶ If so, we should include the liquidation or salvage value as a cash flow
 - ▶ Include the tax effects of the gain or loss that occurs if the actual sale price is larger (smaller) than the book value (original cost net of depreciation)

$$\text{Tax on asset sale} = t \times (\text{Liquidation Value} - \text{Book Value})$$

Incremental Cash Flows Example

- ▶ Which of the following should be treated as incremental cash flows when deciding whether to invest in a new manufacturing plant? The site is already owned by the company, but existing buildings would need to be demolished.
 1. The market value of the site and existing buildings
 2. Demolition costs and site clearance
 3. The cost of a new access road put in last year
 4. Lost earnings on other products from executive time spent on new plant
 5. A fraction of the cost of leasing the president's jet airplane
 6. Future depreciation of the new plant
 7. The reduction in corporate taxes from depreciation of the new plant
 8. The initial investment in inventories of raw materials

Mini Case: IM&C's Garden Fertilizer Project

- ▶ IM&C is considering a proposal to market a new garden fertilizer. The following projections were made (all numbers are adjusted for inflation).
 - ▶ The project requires an investment of \$12M in plant and machinery
 - ▶ The fixed assets will be fully depreciated over 6 years using straight-line depreciation
 - ▶ The plant and machinery can be dismantled and sold at pre-tax salvage value \$1,949,000 in Year 7
 - ▶ The projected sales are (\$ thousands)

Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
\$523	\$12,887	\$32,610	\$48,901	\$35,834	\$19,717

Mini Case: IM&C's Garden Fertilizer Project

- ▶ The projected cost of goods sold (excluding depreciation) is \$837,000 in Year 1 and 60% of sales in Years 2-6
- ▶ Projected SG&A expenses are \$1.1M in Year 1 and grow at 10% in Years 2-6
- ▶ The firm incurs start-up costs of \$4M in Year 0 and \$1.1M in Year 1
- ▶ Project operations require operating net working capital of \$550,000 in Year 1, and 10% of sales in Years 2-6
- ▶ The relevant cost of capital (discount rate) for the Garden Fertilizer project has been estimated to be 20%
- ▶ Should IM&C undertake the new project?

Mini Case: IM&C's Garden Fertilizer Project

► Let's start by projecting the balance sheet items

(\$ thousands)	0	1	2	3	4	5	6	7
Capital Expenditures	12,000							
Gross PPE	12,000	12,000	12,000	12,000	12,000	12,000	12,000	0
– Accumulated Deprec.		2,000	4,000	6,000	8,000	10,000	12,000	
= Book Value (Net PPE)	12,000	10,000	8,000	6,000	4,000	2,000	0	
Asset Sales								1,949
Net Working Capital	0	550	1,289	3,261	4,890	3,583	1,972	0
Change in NWC		550	739	1,972	1,629	(1,307)	(1,612)	(1,972)

See “Session_12_Fall2026.xlsx”

Mini Case: IM&C's Garden Fertilizer Project

► Now the income statement items:

(\$ thousands)	0	1	2	3	4	5	6	7
Sales		523	12,887	32,610	48,901	35,834	19,717	
- Cost of Goods Sold		837	7,732	19,566	29,341	21,500	11,830	
- SG&A		1,100	1,210	1,331	1,464	1,611	1,772	
- Other Cash Costs	4,000	1,100						
+ Asset Sale Profit								1,949
= EBITDA	(4,000)	(2,514)	3,945	11,713	18,096	12,723	6,115	
- Depreciation		2,000	2,000	2,000	2,000	2,000	2,000	
= Operating Profit (EBIT)	(4,000)	(4,514)	1,945	9,713	16,096	10,723	4,115	
Taxes (@ 21%)	(840)	(948)	408	2,040	3,380	2,252	864	409

Mini Case: IM&C's Garden Fertilizer Project

- Combine to derive the free cash flows

(\$ thousands)	0	1	2	3	4	5	6	7
Operating Profit	(4,000)	(4,514)	1,945	9,713	16,096	10,723	4,115	
Profit from Asset Sales								1,949
- Taxes	840	948	(408)	(2,040)	(3,380)	(2,252)	(864)	(409)
+ Depreciation		2,000	2,000	2,000	2,000	2,000	2,000	
- CapEx	(12,000)	0	0	0	0	0	0	0
- Change in NWC	0	(550)	(739)	(1,972)	(1,629)	1,307	1,612	1,972
= Free Cash Flow	(15,160)	(2,116)	2,798	7,701	13,087	11,778	6,863	3,511

Mini Case: IM&C's Garden Fertilizer Project

- ▶ With cash flows in hand, we can compute decision criteria
- ▶ Should IM&C go ahead with the project?

(\$ thousands)	0	1	2	3	4	5	6	7
Free Cash Flow	(15,160)	(2,116)	2,798	7,701	13,087	11,778	6,863	3,511
PV (R = 20%)	(15,160)	(1,763)	1,943	4,457	6,311	4,733	2,298	980
Net Present Value	3,799							
Internal Rate of Return	26.16%							
Payback Period	3.52 Years							
Disc. Payback Period	4.89 Years							

Mini Case: IM&C's Garden Fertilizer Project

- ▶ What would cash flows look like under accelerated depreciation?
- ▶ Let's redo the evaluation with full bonus depreciation

(\$ thousands)	0	1	2	3	4	5	6	7
Operating Profit	(16,000)	(2,514)	3,945	11,713	18,096	12,723	6,115	1,949
- Taxes	(3,360)	(528)	828	(2,460)	(3,800)	(2,672)	(1,284)	(409)
+ Depreciation	12,000	0	0	0	0	0	0	
- CapEx	(12,000)	0	0	0	0	0	0	0
- Change in NWC	0	(550)	(739)	(1,972)	(1,629)	1,307	1,612	1,972
= Free Cash Flow	(12,640)	(2,536)	2,378	7,281	12,667	11,358	6,443	3,511

Mini Case: IM&C's Garden Fertilizer Project

- ▶ Does accelerated depreciation make the project more or less appealing? Why?

(\$ thousands)	0	1	2	3	4	5	6	7
Free Cash Flow	(12,640)	(2,536)	2,378	7,281	12,667	11,358	6,443	3,511
PV (R = 20%)	(12,640)	(2,113)	1,651	4,214	6,109	4,565	2,158	980
Net Present Value	4,922							
Internal Rate of Return	28.90%							
Payback Period	3.44 Years							
Disc. Payback Period	4.61 Years							

Practical Issue

- ▶ Good investment decisions depend both on a sensible decision criterion and sensible forecasts (inputs)
- ▶ Always use a DCF-based capital budgeting technique
 - ▶ Use NPV, IRR, and/or refinements of NPV in case of project interactions
- ▶ Pay close attention to the plausibility of the economic forecasts for the project and try to minimize forecasting errors
 - ▶ An NPV calculation is only as good as its inputs (“GIGO”)
 - ▶ An adjusted discount rate can compensate for economic risk (the fact that future events have not resolved yet) but not biased forecasts

Conclusion

- ▶ Undertake a project only if the NPV of the incremental cash flows is positive
 - ▶ Incremental cash flows: difference with vs without project
- ▶ Incremental cash flows should:
 - ▶ **Include** side benefits and costs, opportunity costs, effect of closure on cash flows
 - ▶ **Ignore** sunk costs, allocated overhead unaffected by the project, cash flow streams unaffected by the project
- ▶ Decisions are only as good as the forecasts behind them!